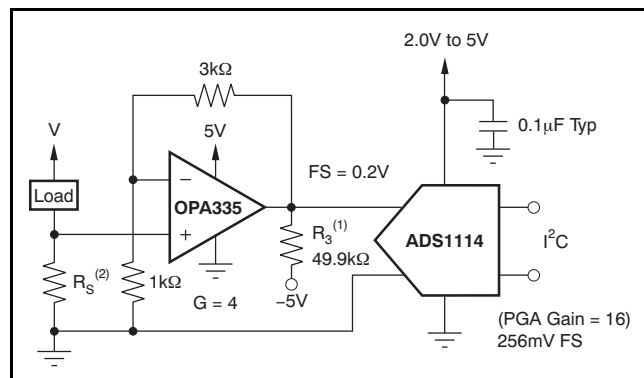


LOW-SIDE CURRENT MONITOR

Figure 38 shows a circuit for a low-side shunt-type current monitor. The circuit monitors the voltage across a shunt resistor, which is sized as small as possible while giving a measurable output voltage. This voltage is amplified by an OPA335 low-drift op amp, and the result is read by the ADS1114/5.

It is suggested that the ADS1114/5 be operated at a gain of 8. The gain of the OPA335 can then be set lower. For a gain of 16, the op amp should be set up to give a maximum output voltage no greater than 0.256V. If the shunt resistor is sized to provide a maximum voltage drop of 50mV at full-scale current, the full-scale input to the ADS1114/5 is 0.2V.



- (1) Pull-down resistor to allow accurate swing to 0V.
- (2) R_S is sized for a 50mV drop at full-scale current.

Figure 38. Low-Side Current Measurement

The ADS1113/4/5 are fabricated in a small-geometry, low-voltage process. The analog inputs feature protection diodes to the supply rails. However, the current-handling ability of these diodes is limited, and the ADS1113/4/5 can be permanently damaged by analog input voltages that remain more than approximately 300mV beyond the rails for extended periods. One way to protect against overvoltage is to place current-limiting resistors on the input lines. The ADS1113/4/5 analog inputs can withstand momentary currents as large as 100mA.

If the ADS1113/4/5 are driven by an op amp with high-voltage supplies, such as $\pm 12V$, protection should be provided, even if the op amp is configured so that it does not output out-of-range voltages. Many op amps drift to one of the supply rails immediately when power is applied, usually before the input has stabilized; this momentary spike can damage the ADS1113/4/5. This incremental damage results in slow, long-term failure, which can be disastrous for permanently installed, low-maintenance systems.

If an op amp or other front-end circuitry is used with an ADS1113/4/5, performance characteristics must be taken into account when designing the application.

PACKAGING INFORMATION

Orderable Device	Status ⁽¹⁾	Package Type	Package Drawing	Pins	Package Qty	Eco Plan ⁽²⁾	Lead/Ball Finish	MSL Peak Temp ⁽³⁾
ADS1113IDGSR	PREVIEW	MSOP	DGS	10	2500	TBD	Call TI	Call TI
ADS1113IDGST	PREVIEW	MSOP	DGS	10	250	TBD	Call TI	Call TI
ADS1113IRUGR	PREVIEW	QFN	RUG	10	3000	TBD	Call TI	Call TI
ADS1113IRUGT	PREVIEW	QFN	RUG	10	250	TBD	Call TI	Call TI
ADS1114IDGSR	PREVIEW	MSOP	DGS	10	2500	TBD	Call TI	Call TI
ADS1114IDGST	PREVIEW	MSOP	DGS	10	250	TBD	Call TI	Call TI
ADS1114IRUGR	PREVIEW	QFN	RUG	10	3000	TBD	Call TI	Call TI
ADS1114IRUGT	PREVIEW	QFN	RUG	10	250	TBD	Call TI	Call TI
ADS1115IDGSR	PREVIEW	MSOP	DGS	10	2500	TBD	Call TI	Call TI
ADS1115IDGST	PREVIEW	MSOP	DGS	10	250	TBD	Call TI	Call TI
ADS1115IRUGR	PREVIEW	QFN	RUG	10	3000	TBD	Call TI	Call TI
ADS1115IRUGT	PREVIEW	QFN	RUG	10	250	TBD	Call TI	Call TI

⁽¹⁾ The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

⁽²⁾ Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check <http://www.ti.com/productcontent> for the latest availability information and additional product content details.

TBD: The Pb-Free/Green conversion plan has not been defined.

Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.

Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.

Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

⁽³⁾ MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

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